GPTfy POC1 - Project Walkthrough

Audience: Team / client presentation
Prepared by: Project creator
Date: June 5, 2026
Org: gptfy-poc1 (kesavpoc@gptfy.com)
Latest checkpoint: Git tag Part-1 · latest commit 4a5a27a

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1. Why this project exists

Client: Tungsten Automation
Product: GPTfy (AI layer on Salesforce)
POC scope: Use Case 2 - Customer Assist + Knowledge Base Creation via Portal

The business problem is straightforward:

- Customers wait too long for simple answers.
- Agents spend time on questions that could be self-served.
- When cases are resolved, that knowledge is rarely captured in a reusable Knowledge Base.

What we built (Part 1 complete + Part 2 in progress):

Goal	How we achieve it
Deflect simple support questions with AI	Guest portal + GPTfy RAG (two AI paths available)
Close cases automatically when the customer confirms	Platform Event + trigger
Let agents resolve in one click	Resolve quick action
Build a KB over time	Draft Knowledge articles on agent resolve + KB data pipeline
Email the answer to the customer	Record-triggered email flow
Support multiple Tungsten products	E-Invoicing, TotalAgility, PowerPDF

POC products: E-Invoicing, TotalAgility, PowerPDF (3-week POC, $9,999 investment per signed scope doc).

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2. Who uses the system

Person	Role
Guest (portal visitor)	Submits a question, reads the AI answer, confirms or escalates
Support agent	Uses the Resolve button, reviews draft KB articles
Admin / developer	Deploys metadata, configures flows/triggers, monitors email

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3. The big picture - three user paths, two AI modes

Everything runs on Salesforce in org gptfy-poc1, with GPTfy (ccai managed package) for AI.

Three user-facing paths

Path 1 - Guest Portal: Guest on Experience Cloud site (gptfysupport1) -> caseResolutionAssistant LWC -> AI fills Case.Description -> Platform Event closes Case.

Path 2 - Agent Resolve: Support Agent -> Resolve Quick Action -> Draft Knowledge Article created.

Path 3 - Resolution Email: Any case close (guest or agent) -> Case_Resolution_Email_RTF -> Email to SuppliedEmail.

One rule ties it together: whenever a Case closes with a resolution and a guest email, the customer gets that answer by email.

Two AI resolution modes (switchable, never both active)

Mode	Trigger	How it works
PATH 1 - Flow/GCP	Case_Resolution_RTF record-triggered flow (active)	Invokes ccai__AIPromptProcessingInvokable via GPT
PATH 2 - Agent/RAG	CaseAIAgentTrigger Apex trigger (active)	Enqueues CaseAgentResolutionService (Queueable

Both modes write to the same field. The LWC polling loop, Platform Event close path, and email flow are shared and path-agnostic.

Switch rule: Activate exactly one. If both are active, two AI calls race to write Case.Description.

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4. Guest Portal path (step by step)

This is the self-service deflection path. A customer visits the public Experience Cloud site (gptfysupport1).

Step 1 - Customer fills the form

The UI is the LWC caseResolutionAssistant, embedded on the site home page.

Fields:

- First name, last name, email (required)
- Product (dynamic picklist - values loaded live from Case.Product__c via getProductOptions())
- Question (required, max 255 chars)

The LWC validates input (email regex, required fields) before submit.

Current POC products in picklist: E-Invoicing, TotalAgility, PowerPDF.

Step 2 - Case is created

On Find Solution, the LWC calls Apex: CaseResolutionController.createSupportCase(...)

This creates a Case with:

- Origin = 'Web' (critical - gates both AI modes)
- Status = 'New'
- Subject = the question (truncated to 255 chars)

- SuppliedName, SuppliedEmail = guest info
- Product__c = selected product (optional; null means search all products)

Why without sharing? Guest users on Experience Cloud cannot normally create Cases under default sharing. This controller runs elevated for these three narrow, guest-safe operations only.

Step 3 - AI generates the answer

PATH 1 (flow): Case insert fires Case_Resolution_RTF -> invokes GPTfy -> writes answer to Case.Description.

PATH 2 (agent): Case insert fires CaseAIAgentTrigger -> enqueues CaseAgentResolutionService -> calls ccai.AIAgenticUtility.invokeAgent() with the cleaned subject -> strips HTML from response -> writes plain-text answer to Case.Description.

Subject cleaning (PATH 2): surrounding quotes are stripped before the RAG query (e.g. "What is TotalAgility?" -> What is TotalAgility?) because they hurt vector similarity scoring.

The LWC shows a loading spinner and polls every 2.5 seconds, up to 36 times (90 seconds total) via CaseResolutionController.getRecommendation(caseId).

- When Description is populated -> pass through sanitizeRecommendation() -> show the answer.
- If still empty after 90 s -> show TIMEOUT_MESSAGE: "Our support team has your case and will follow up with you shortly."

Step 4 - Low-quality response filtering

Before the answer is shown to the guest, sanitizeRecommendation() checks for known junk/confusion responses from the AI (e.g. "I didn't get that", "no answer found in the knowledge base", "can you repeat", etc.).

- Exact match - normalised lowercase against a LOW_QUALITY_RESPONSES set
- Contains match - substring scan against a LOW_QUALITY_CONTAINS list (catches long fallback sentences in agent system prompts)

If junk is detected -> show NO_ANSWER_MESSAGE: "We could not find a specific answer in our knowledge base at this time. Our support team will review your case and follow up with you shortly."

This means guests never see raw AI error text.

Step 5 - Customer chooses

Two buttons:

Button	What happens
Yes, resolved	Case closes; resolution saved; email sent
No, create a case	Case stays open; customer sees case number for agent follow-up

Step 6 - "Yes, resolved" (the close path)

Guest users cannot update Cases directly (Guest User License restriction - not bypassed by without sharing).

So we use a Platform Event pattern:

1. LWC calls resolveCase(caseId)
2. Apex publishes Case_Resolved__e with the Case Id
3. CaseResolvedTrigger runs as Automated Process (system context):
- Sets Status = 'Closed'
- Copies Description -> Resolution__c (what the customer saw)
4. Case update fires the email flow (Path 3 below)

The LWC shows "Closing your case..." while this completes (usually 1-2 seconds).

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5. Agent Resolve path (step by step)

When AI cannot help, or the customer clicks No, the case stays open for an agent.

Step 1 - Agent opens an open Case

On the Case record page (highlights panel), the agent sees a Resolve button - only when Status != Closed (Dynamic Action visibility filter; one-time Setup step documented in docs/resolve_button_setup.md).

Step 2 - Agent clicks Resolve

A modal opens (LWC caseResolveAction, wrapped in Aura caseResolveActionAura because of a known Salesforce quirk with LWC quick actions on Case).

The agent types the resolution in a rich-text area.

Step 3 - Agent clicks Save

Apex CaseResolveActionController.saveResolution(caseId, resolutionText) does three things:

1. Updates the Case: Resolution__c = typed text, Status = 'Closed'
2. Creates or updates a Draft Knowledge article:
- Title = Case Subject
- Resolution = agent's text
- Linked to Case via CaseArticle
- Dedupes: if a draft already exists for this case, it updates instead of creating a duplicate
3. Returns a result so the LWC can toast: "Case resolved. Draft Knowledge article created/updated."

KB creation is best-effort - if it fails, the case still closes and a warning toast is shown.

Step 4 - Email (if guest email exists)

Same email flow as the portal path (Path 3 below).

Design choice: KB drafting is triggered on agent resolve, not on guest self-resolve. Portal-deflected closes copy AI text to Resolution__c but do not auto-draft KB articles yet (KCS-compliant KB drafting prompt is still pending).

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6. Resolution email (automatic)

Flow: Case_Resolution_Email_RTF (Active)

When it runs:

- Case is updated to Status = Closed
- SuppliedEmail is not blank
- Resolution__c is not blank

What the customer receives:

Field	Source
To	Case.SuppliedEmail
Subject	Case.Subject (their original question)
Body	Intro + decoded resolution text
From	Org-wide email kesavamoorthy@cloudcompliance.app

Technical detail: Resolution__c is an HTML field, so quotes can store as ". The flow decodes HTML entities before sending so the email looks correct.

Note on sender domain: Email currently sends via ...@sfcustomeremail.com until DKIM is verified for cloudcompliance.app.

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7. Knowledge Base data pipeline (NEW in Part 2)

Tungsten product FAQ content has been scraped, cleaned, and loaded into Salesforce Knowledge to power the RAG vector store.

Data loaded

Product	Source file	Approx. rows
E-Invoicing	data/einvoicing-faq-kb.csv	~2,600
PowerPDF	data/powerpdf-faq-kb.csv	~5,100
TotalAgility	data/totalagility-faq-kb.csv	~78,000

How the pipeline works

```
Tungsten support site
    -> scripts/scrape_tungsten_kb.py        (Python scraper)
    -> data/*-faq-kb.csv                    (cleaned CSV)
    -> scripts/batch_import_kb.py           (batch REST import)
    OR
    -> scripts/generate_kb_apex.py          (generates Apex batch scripts)
    -> scripts/apex/batches/batch_001.apex  (executed via sf CLI)
    -> Salesforce Knowledge articles        (in gptfy-poc1 org)
    -> GPTfy vector store index             (powers RAG for PATH 2)
```

Why this matters

The vector store is only as good as the KB content loaded into it. With ~85,000 FAQ rows across three products, PATH 2 (Agent/RAG) has a rich source to query against.

8. Key Salesforce fields (Case)

Field	Purpose
Subject	Guest question / email subject
Description	AI-generated answer (written by PATH 1 or PATH 2)
Resolution__c	Final resolution text (Html field; copied from Description on close)
SuppliedEmail	Guest email for notifications
SuppliedName	Guest name from form
Origin	Web for portal cases - gates both AI modes
Product__c	Product picklist - E-Invoicing / TotalAgility / PowerPDF
Status	Closed triggers email flow

Many other GPTfy-related fields exist on Case (sentiment, RCA, triage, etc.) for Use Case 1 (Case Intelligence) - provisioned in schema but not in current focus.

9. What's in the repo (technical inventory)

```
force-app/main/default/
??? lwc/
?    ??? caseResolutionAssistant/         ? Guest portal UI (dynamic picklist, 90s poll, response
sanitization)
?   ??? caseResolveAction/            ? Agent resolve modal
??? aura/
?   ??? caseResolveActionAura/        ? Wrapper for quick action
??? classes/
?     ??? CaseResolutionController.cls     ? Guest Apex (without sharing; sanitizeRecommendation;
getProductOptions)
?   ??? CaseResolveActionController.cls ? Agent Apex (with sharing)
?   ??? *Test.cls                     ? Unit tests
??? triggers/
?   ??? CaseResolvedTrigger.trigger    ? Platform event subscriber
??? objects/
?   ??? Case_Resolved__e/             ? Platform event
?   ??? Case/fields/                  ? Custom fields: Resolution__c, Product__c
??? flows/
?   ??? Case_Resolution_RTF.flow       ? PATH 1: GPTfy AI prompt flow
?   ??? Case_Resolution_Email_RTF.flow ? Resolution email
??? quickActions/
?   ??? Case.Resolve                  ? Agent quick action
??? permissionsets/
?   ??? gptfysupport_Guest_Access      ? Minimal guest permissions
??? settings/
```

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?   ??? EmailAuthorization              ? Substitute sender workaround
??? digitalExperiences/site/
    ??? gptfysupport1/                  ? Experience Cloud site source

force-app-agent-path/main/default/
??? triggers/
?   ??? CaseAIAgentTrigger.trigger     ? PATH 2: fires on Case insert (Origin=Web)
??? classes/
    ??? CaseAgentResolutionService.cls ? PATH 2: Queueable; calls ccai.AIAgenticUtility.invokeAgent()

data/
??? einvoicing-faq-kb.csv               ? E-Invoicing KB (~2,600 rows)
??? powerpdf-faq-kb.csv                 ? PowerPDF KB (~5,100 rows)
??? totalagility-faq-kb.csv             ? TotalAgility KB (~78,000 rows)

scripts/
??? scrape_tungsten_kb.py               ? Scrapes Tungsten support site
??? batch_import_kb.py                  ? Batch-imports KB CSV via REST API
??? generate_kb_apex.py                 ? Generates Apex batch scripts from CSV
??? generate_kb_update_apex.py          ? Generates Apex update scripts
??? apex/
    ??? agent-path/testAgentPath.apex  ? PATH 2 smoke test
    ??? testRecommendationSanitize.apex
    ??? testIrrelevantQuestion*.apex   ? RAG quality tests
    ??? testEssentialEightRag.apex
    ??? testProductPicklist.apex
    ??? testConfigUnblockers.apex
    ??? testCaseResolutionEmail.apex
    ??? testCaseResolveActionUI.apex
    ??? testCaseResolveKbDraft.apex
    ??? describeAgent.apex              ? Agent inspection utilities
    ??? readAgentPrompt.apex
    ??? updateAgentPrompt.apex
    ??? verifyAgentPrompt.apex

docs/
??? PRD.md                              ? Living product doc (auto-changelog)
??? GPTfy-POC1-Project-Walkthrough.md ? This file

POC_CHECKLIST.md                        ? Full POC tracker
```

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10. What's done vs. what's still pending

Done - Part 1 (verified in org, 2026-06-03)

- Guest portal LWC + Apex + polling
- Platform Event close path for guests
- Agent Resolve quick action + KB draft
- Resolution email with plain-text body fix
- Apex unit tests + org verification scripts
- GitHub repo + Part 1 checkpoint (Part-1 tag)

Done - Part 2 (in progress, 2026-06-04 -> 2026-06-05)

- PATH 2: CaseAIAgentTrigger + CaseAgentResolutionService (Queueable, callout to ccai.AIAgenticUtility)
- Switch mechanism between PATH 1 and PATH 2
- Low-quality AI response filtering (sanitizeRecommendation, 30+ junk patterns)
- Dynamic product picklist (getProductOptions) - E-Invoicing, TotalAgility, PowerPDF
- cleanSubject() - strips surrounding quotes before RAG query
- stripHtml() - strips HTML tags and collapses blank lines in agent response
- KB data files loaded: ~85,000 FAQ rows across 3 products
- KB import/scrape pipeline (Python + Apex)
- Experience Cloud site source (gptfysupport1)
- Extended polling window: 30 s -> 90 s
- Expanded verification scripts (RAG quality, sanitize, config, product, agent path)
- Agent prompt utility scripts (describe, read, update, verify)

Still pending (for full POC sign-off)

Priority	Item
High	Decide AI path for demo - activate PATH 1 (flow) or PATH 2 (agent) - neve
High	Complete TotalAgility KB import (large dataset - batch jobs)
Medium	GPTfy KCS-compliant KB drafting prompt for agent resolve
Medium	DKIM verification for cloudcompliance.app email domain
Medium	Deflection metrics and reports
Medium	UAT + demo recording for leadership
Low	Finalize Experience Cloud site branding (gptfysupport1)

Out of scope for now: Use Case 1 (Case Intelligence) and Use Case 3 (Microsoft Copilot).

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11. How to demo this to your team/client

Pre-demo checklist

- [ ] Confirm which AI path is active (check Setup -> Apex Triggers for CaseAIAgentTrigger, and Flows for Case_Resolution_RTF)
- [ ] Verify KB articles are published in org (at least PowerPDF and E-Invoicing)
- [ ] Open Experience Cloud site URL in incognito before presenting

Demo script - Guest portal path

1. Open the public gptfysupport1 Experience Cloud site in incognito
2. Fill in: first name, last name, email, select a product (e.g. PowerPDF), type a question
3. Click Find Solution - wait for AI answer (~5-30 seconds depending on active path)
4. Show the AI answer on screen
5. Click Yes, resolved

6. In Salesforce: show Case status = Closed, Resolution__c populated, email received

Demo script - Agent resolve path

1. Open an open Case (or one where the guest clicked No)
2. Click Resolve in the highlights panel -> type a resolution -> Save
3. Show: Case closed, toast confirms "Draft Knowledge article created/updated"
4. Open the linked Knowledge article in draft status
5. If SuppliedEmail exists on the case, show the email received by the customer

Verification in org

```
# Smoke test all paths
sf apex run --file scripts/apex/testCaseResolutionEmail.apex --target-org gptfy-poc1

# Test PATH 2 (Agent/RAG) specifically
sf apex run --file scripts/apex/agent-path/testAgentPath.apex --target-org gptfy-poc1

# Test RAG answer quality
sf apex run --file scripts/apex/testEssentialEightRag.apex --target-org gptfy-poc1
```

Test records use subject prefix cursor-test-{timestamp} for easy cleanup.

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12. One-sentence pitch for leadership

GPTfy POC1 lets customers get instant AI answers on a public portal and close their own cases when satisfied - while agents resolve complex cases in one click and automatically draft Knowledge articles - with every resolution emailed back to the customer - backed by 85,000 Tungsten product FAQ articles in the vector store.

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13. Suggested talking order for your presentation

1. Business problem (Section 1) - 2 min
2. Three paths + two AI modes overview (Section 3) - 3 min
3. Live demo: guest portal (Section 4) - 5 min
4. Live demo: agent resolve + KB (Section 5) - 3 min
5. Email confirmation (Section 6) - 1 min
6. KB data pipeline (Section 7) - 2 min
7. What's next (Section 10) - 2 min

This document is maintained in source control at docs/GPTfy-POC1-Project-Walkthrough.md.